Predicting Stroke Occurrence Using Cross-Validated Supervised and Unsupervised Machine Learning

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***Abstract*—Predicting strokes is highly critical for modern healthcare and early medical intervention. This project applies a rigorous machine learning pipeline to a highly imbalanced dataset of 5,109 patient records to identify physiological and lifestyle factors driving stroke risk. We refined the dataset through extensive Exploratory Data Analysis (EDA) and pre-processing, encoding categorical features into binary data using mapping and One Hot Encoding, utilizing Pearson and Spearman Rank Correlation to evaluate linear and monotonic relationships among encoded categorical and continuous features. To resolve severe class imbalance and overcome a deceptive 95.1% Majority-Class baseline, we implemented RandomUnderSampling. We evaluated three supervised models: Logistic Regression, K-Nearest Neighbors (KNN), and a Neural Network alongside an Unsupervised K-Means Clustering model. Utilizing Stratified 5-Fold Cross-Validation, Logistic Regression performed the best overall with one of the highest recall (0.8753) and the highest AUC-ROC (0.8380) and PR-AUC score (0.1955). We ranked the Neural Network the second best model due to its lower AUC-ROC (0.8319) and PR-AUC score (0.1855). The KNN model achieved the highest Recall (0.8843) but falls behind in the other metrics. The K-Means clustering model did not perform as well as the others for this dataset. We focus on Recall and PR-AUC score because they are significant metrics for identifying class separation and true stroke risks compared to other metrics like Precision and AUC-ROC score. This report details the methodology, feature engineering, training challenges, and evaluation metrics.**

***Index Terms*—Stroke Prediction, Machine Learning, K-Means Clustering, Neural Network, Logistic Regression, KNN, Classification.**

I. Introduction

STROKE is a leading cause of mortality and long-term disability globally. Early identification of high-risk patients allows medical professionals to implement preventative care, significantly reducing healthcare burdens and saving lives. By leveraging machine learning on historical patient data—encompassing demographics, pre-existing conditions like heart disease and hypertension, and lifestyle metrics like smoking status—we aim to accurately predict the discrete binary target of "Stroke" (0 for no stroke, 1 for stroke).

Because stroke events represent a very small minority of the general population, building robust predictive models requires navigating severe class imbalance. If models are evaluated purely on standard accuracy, they appear highly performant by simply predicting the majority class (no stroke) every time. Therefore, this project implements a rigorous evaluation methodology combined with advanced pre-processing techniques, to ensure reliable, real-world applicability that prioritizes high recall and strong class separation.

II. Dataset Description

The utilized dataset tracks various medical and lifestyle metrics to predict stroke probability.

Dataset Size & Cleaning: The raw dataset consisted of 5,110 patient records with 12 original features. During initial EDA, we removed the non-predictive 'id' column and dropped a single anomaly row where gender was classified as 'Other', resulting in a finalized working dataset of 5,109 records.

Class Distribution: The dataset exhibits a severe class imbalance. Out of 5,109 patients, 4,861 (95.1%) did not suffer a stroke, while only 249 (4.9%) did. Because of this, a simplistic algorithm guessing "No Stroke" every single time would inherently be 95.1% accurate without learning any underlying patterns. In machine learning, this is known as the **Majority Class Baseline** or **Zero-R Baseline**. Without proper intervention, models fall victim to the "Accuracy Paradox," achieving this high baseline accuracy while completely failing to identify actual stroke victims.

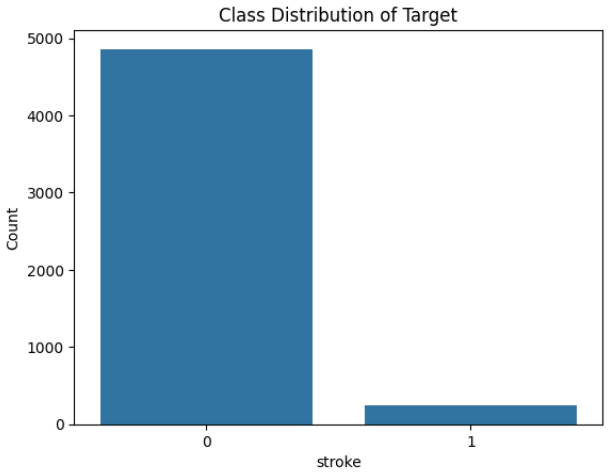


Fig. 1. Distribution of the target variable (stroke), demonstrating a severe 95:5 class imbalance.

III. Exploratory Data Analysis & Pre-processing

Thorough pre-processing is the foundation of any reliable machine learning pipeline. Failing to properly clean, scale, and balance data leads to artificial bias, runtime errors, and poor generalization on unseen data.

A. Handling Null and Duplicate Values

Raw datasets often contain missing data or duplicated rows. We verified the dataset contained 0 duplicate records. However, the 'bmi' column contained 201 missing (NaN) values.

Because BMI is a critical continuous health metric, dropping these rows would result in unacceptable data loss. But later, the 'bmi' column was dropped during feature selection. So no imputation was required.

B. Encoding Categorical Values

Machine learning models require numerical tensors to process data. We applied binary mapping for features with two states (`gender`, `ever_married`, `Residence_type`). For nominal categories with multiple states (`work_type` and `smoking_status`), we utilized one-hot encoding (`pd.get_dummies`), expanding the dataset to 18 mathematically digestible vectors.

C. Correlation Analysis: Spearman and Pearson

To understand relationships and detect potential multicollinearity, we generated a Spearman Rank Correlation Heatmap for continuous variables (`age`, `avg_glucose_level`, `bmi`). Spearman correlation (ρ) evaluates monotonic relationships based on ranked values (d_i being the difference in ranks):

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \tag{1}$$

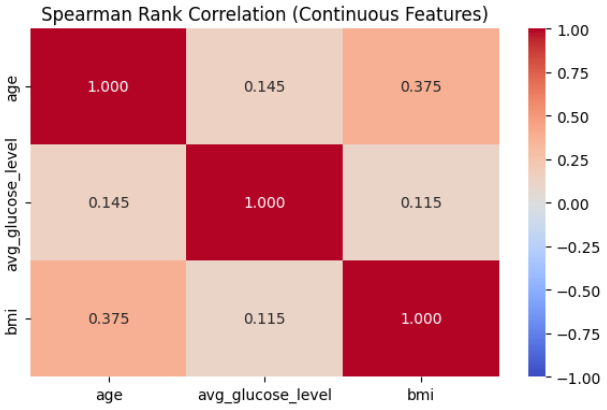


Fig. 2. Spearman Rank Correlation matrix assessing the monotonic relationships between continuous health metrics.

We observed moderate monotonic correlation between age and bmi (0.375). We then evaluated linear relationships across the fully encoded dataset using the Pearson correlation coefficient (r):

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{2}$$

We drew a Pearson Rank Correlation Heatmap showing correlation among all features. This confirmed strong predictive signals from features like `age` against the `stroke` target, guiding our model's weight adjustments. We also saw strong collinearity between age and ever_married. And we also saw that the one-hot encoded work_type features had very low correlation to the target feature while having high collinearity with age.

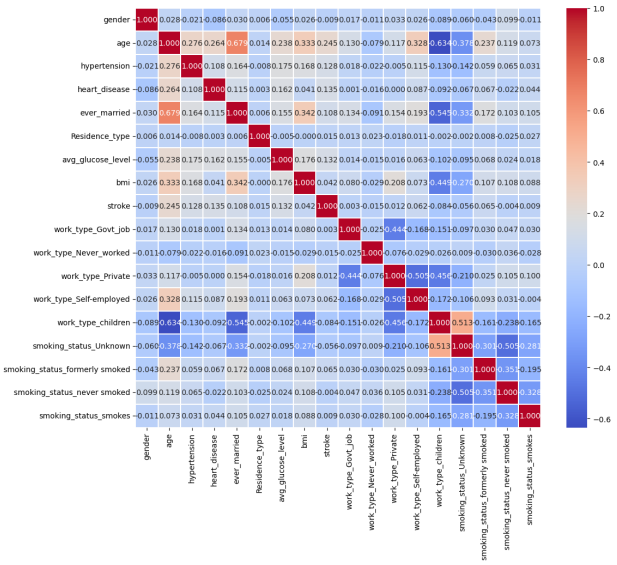


Fig. 3. Pearson Rank Correlation matrix assessing the linear relationships between continuous and categorical features.

D. Feature Selection

We removed the 'gender' column due to its low correlation with 'stroke', and removed the 'ever_married' and 'bmi' column due to their high collinearity with 'age'. We also removed all the 'work_type' columns due to their low correlation to the target and the 'smoking_status_Unknown' column because it is simply an absence of information. Keeping it trains the model to predict patterns from it.

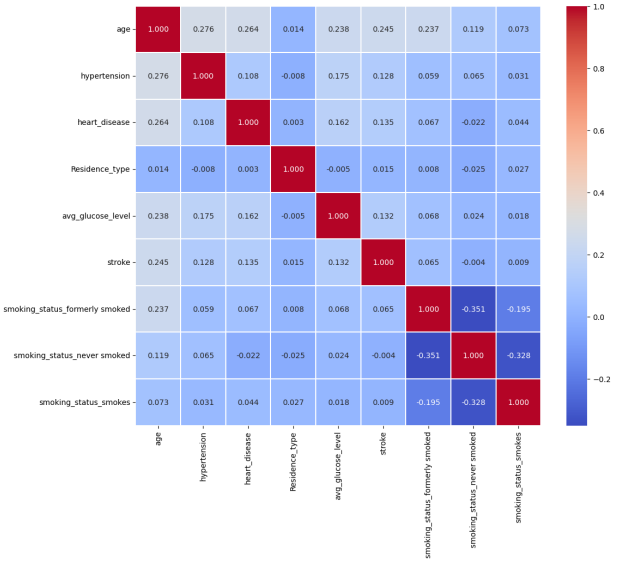


Fig. 4. Pearson Rank Correlation matrix after optimized feature selection.

E. Feature Scaling

Our continuous features spanned vastly different ranges (e.g., `avg_glucose_level` from 55 to 271 vs. encoded binary categories of 0 and 1). Without standardizing these values, algorithms like KNN and Neural Networks would treat

large-magnitude features as artificially more important. We applied a `StandardScaler` to force a mean (μ) of 0 and a standard deviation (σ) of 1:

$$z = \frac{x - \mu}{\sigma} \tag{3}$$

F. Class Imbalance: RandomUnderSampling

To solve the Accuracy Paradox and overcome the 95.1% Zero-R baseline, we utilized RandomUnderSampling. SMOTE draws lines between minority class points to create synthetic data. When working with a high-dimensional dataset filled with weak predictors, SMOTE generates synthetic stroke cases that are mathematically blurry across those weak dimensions. This softens the decision boundary and causes the model to miss actual borderline stroke cases, lowering recall. By using undersampling on a heavily pruned, high-signal feature space (Age, Glucose, Heart Disease, Hypertension), we force the models to draw strict decision boundaries around the real, unadulterated stroke cases.

IV. Model Methodologies and Training Challenges

A. Logistic Regression

Logistic Regression estimates stroke probability by fitting a linear equation to the scaled features and passing it through a logistic function:

$$P(y = 1|x) = \frac{1}{1 + e^{-(w \cdot x + b)}} \tag{4}$$

Training Challenges: Logistic regression assumes a linear relationship between the independent features and the log-odds of the target. If underlying medical interactions are highly non-linear, the model may underfit compared to more flexible architectures like neural networks.

B. Neural Network

We constructed a multi-layer feed-forward Neural Network using the continuous and encoded categorical features as inputs mapping to hidden Dense layers utilizing the Rectified Linear Unit (ReLU) activation function:

$$\text{ReLU}(z) = \max(0, z) \tag{5}$$

The output layer utilizes a sigmoid activation to output a continuous stroke probability:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{6}$$

The loss function is binary cross-entropy:

$$L = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \tag{7}$$

Training Challenges: Undersampling drastically reduces the overall size of the training set by throwing away a vast majority of the majority class (no stroke) samples to match the minority class. This deprives the neural network of abundant background data, making it harder for hidden layers to learn robust, generalized representations of healthy physiological patterns.

C. K-Nearest Neighbors (KNN)

KNN predicts stroke risk by calculating the Euclidean distance between a test patient and historical training data:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2} \tag{8}$$

Training Challenges: KNN suffers heavily from the "Curse of Dimensionality" in datasets with many one-hot encoded categorical features. Distances in high-dimensional space become uniform, causing KNN to struggle with precise boundary definition compared to parametric models.

D. K-Means Clustering (Unsupervised)

K-Means clustering is an unsupervised learning algorithm used to partition the feature space into K distinct clusters without relying on pre-labeled target classes. The algorithm initializes K centroids and iteratively assigns each patient record x_i to the nearest centroid based on squared Euclidean distance:

$$J = \sum_{j=1}^{K} \sum_{i in S_j} ||x_i - \mu_j||^2 \tag{9}$$

where μ_j is the centroid of cluster S_j. Once assigned, centroids are recomputed as the mean of all data points belonging to the cluster until convergence.

Training Challenges: Because K-Means groups data purely by feature variance and distance rather than the target label, clusters do not cleanly align with the binary stroke outcome, resulting in poor predictive metrics like low Recall and AUC-ROC.

V. Cross-Validated Results

A. Evaluation Metrics

Because overall accuracy is inherently deceptive in a dataset where 95.1% of labels belong to a single class, evaluating medical diagnostics relies heavily on Recall (Sensitivity) and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). Recall ensures we minimize False Negatives (critical misses where a stroke-prone patient is cleared).

$$\text{Precision} = \frac{TP}{TP + FP} \tag{10}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{11}$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{12}$$

TABLE I
Aggregated 5-Fold Model Comparison

Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC	PR-AUC
Logistic Regression	0.6721	0.1175	0.8753	0.2070	0.8380	0.1955
Neural Network	0.6851	0.0.1182	0.8429	0.2072	0.8319	0.1855
KNN	0.6310	0.1071	0.8834	0.1906	0.8110	0.1493
K-Means (Unsup.)	0.8444	0.1342	0.4020	0.2012	0.6345	0.0862

B. Confusion Matrices (Fold Averages)

Fig. 5 displays the average confusion matrix per validation fold. In the context of our stroke prediction problem: **True Positives (TP)** are correctly identified stroke instances, **True Negatives (TN)** are correctly identified non-stroke instances, **False Positives (FP)** are false alarms (non-stroke patients predicted to have a stroke), and **False Negatives (FN)** are critical misses (stroke patients predicted not to have a stroke).

- **Logistic Regression:** The confusion matrix shows that the model prioritized minimizing critical misses, correctly identifying 43.6 stroke instances (TP) and 643.2 non-stroke instances (TN) per fold. It kept critical misses down to just 6.2 (FN), though at the cost of 328.8 false alarms (FP).
- **Neural Network:** The confusion matrix shows that the model successfully caught 42.0 stroke instances (TP) and 658.0 non-stroke instances (TN) per fold, managing 314.0 false positives (FP) while keeping critical misses low at 7.8 (FN).
- **KNN:** The confusion matrix shows that the model identified 44.0 stroke instances (TP) and 600.8 non-stroke instances (TN) correctly, but suffered from the highest rate of false alarms, generating 371.2 false positives (FP) per fold while missing 5.8 stroke instances (FN).
- **K-Means:** The confusion matrix shows that because unsupervised clustering groups data purely by distance without leveraging target labels, it struggled to align with the binary stroke outcome, resulting in a high number of critical misses with 29.8 false negatives (FN) per fold, alongside 20.0 true positives (TP), 842.8 true negatives (TN), and 129.2 false positives (FP).

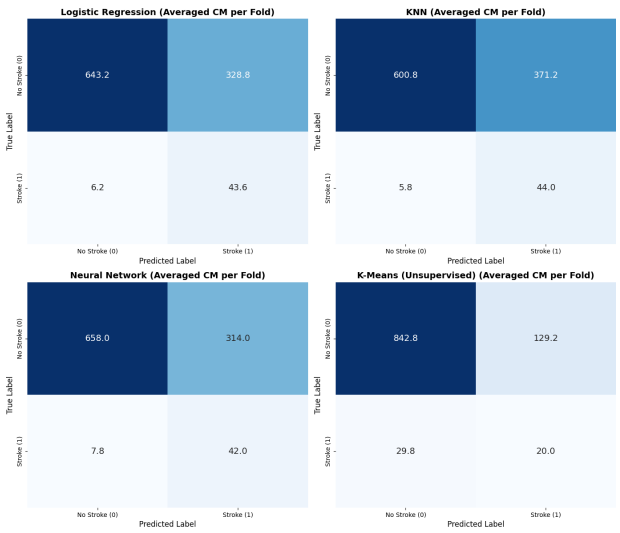


Fig. 5. Averaged Confusion Matrices across the 5 validation folds. Decimal values reflect the mathematical mean of TP, TN, FP, and FN counts per fold.

C. ROC Curves

The ROC curve evaluates the true positive rate versus the false positive rate across all prediction thresholds. Logistic Regression dominates with an aggregated AUC of 0.8380, followed by the Neural Network at 0.8319. KNN has an AUC score of 0.8110 and K-Means has the poorest AUC score of 0.6345.

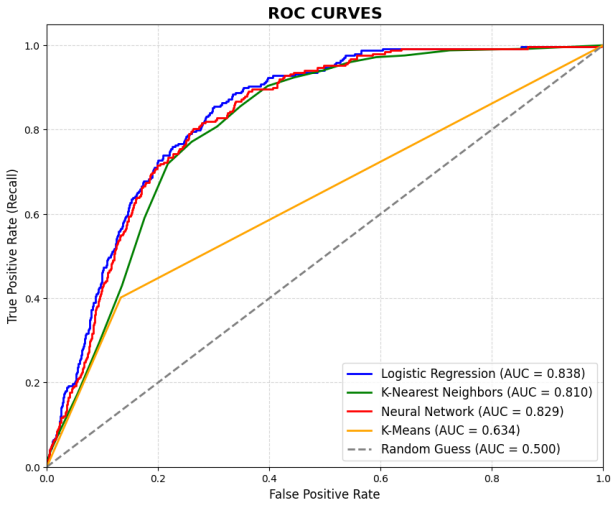


Fig. 6. Cross-validated ROC Curves displaying the superior class separation capabilities of Logistic Regression and the Neural Network.

D. PR Curves

The Precision-Recall (PR) curve evaluates the trade-off between precision and recall across all classification thresholds, which is critical for heavily imbalanced datasets. Logistic Regression leads with an aggregated PR-AUC score of 0.1955, closely followed by the Neural Network at 0.1855. KNN achieves a PR-AUC score of 0.1493, while K-Means performs the poorest with a PR-AUC score of 0.0862 due to unsupervised limitations in minority class isolation.

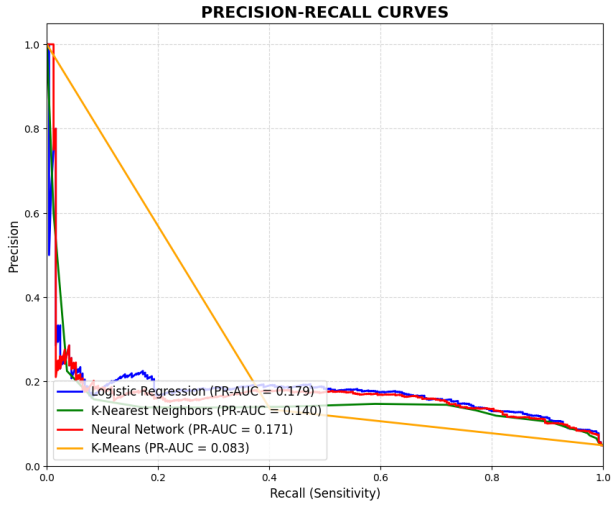


Fig. 7. Cross-validated Precision-Recall Curves demonstrating the performance of each model on the imbalanced stroke prediction dataset.

VI. Conclusion

Predicting medical events like strokes requires meticulous data handling. Without applying Standard Scaling for continuous variables using ColumnTransfer to keep the categorial

features unchanged, and deploying RandomUnderSampler to balance the 95:5 class disparity, machine learning models default to predicting the 95.1% Zero-R baseline.

Based on the comprehensive cross-validated evaluation of our machine learning pipeline, Logistic Regression stands out as the overall best-performing and most reliable model for stroke prediction, closely complemented by the Neural Network.

A. Best Model and Performance Summary

Logistic Regression achieved the highest discriminative power with an aggregated **AUC-ROC of 0.8380** and a **PR-AUC of 0.1955**, alongside a strong **Recall of 0.8753**. While the Neural Network marginally outperformed it in raw accuracy (0.6851 vs. 0.6721) and F1-score (0.2072), Logistic Regression delivered superior class separation and clinical interpretability. Conversely, unsupervised K-Means clustering and distance-based K-Nearest Neighbors struggled significantly due to the high dimensionality of encoded nominal features and the absence of target guidance during unsupervised partitioning.

B. Probable Reasons Behind the Results

The observed performance metrics and model behaviors stem from several key methodological factors:

- **Impact of Undersampling on Model Capacity:** By applying undersampling to address the severe 95:5 class imbalance, we successfully forced models to learn minority class boundaries rather than defaulting to the majority-class baseline. However, this drastically reduced the training set size, introducing higher variance and making deep models prone to overfitting—hence the necessity of early stopping and regularization for the Neural Network.
- **Linear vs. Non-Linear Separability:** Logistic Regression excelled because key predictors like age and average glucose level exhibit strong linear and monotonic relationships with stroke risk. Its parametric nature prevented it from chasing noise in the smaller, undersampled dataset.
- **Curse of Dimensionality in Distance Metrics:** KNN and K-Means relied heavily on Euclidean distance across one-hot encoded lifestyle and categorical features. In high-dimensional sparse spaces, distance metrics degrade, leading to high false-positive rates and lower precision-recall trade-offs.

Ultimately, scaled, regularized parametric models like Logistic Regression and Neural Networks offer the most viable, robust mechanisms for proactive clinical risk assessment when working with constrained, imbalanced medical data.

REFERENCES

[1] Stroke Prediction Dataset. Available: 12.csv.